

Jenkins Automation Server and CI/CD

Description

1. Learn basics of Jenkins automation server.
2. Learn CI/CD deployment method used in Jenkins.
3. Install Jenkins on local machine.

1. Basics of Jenkins Automation Server

What is Jenkins?

Jenkins is an open-source **automation server** used to automate software development processes.

Jenkins can automate:

- Building applications
- Testing applications
- Deploying applications
- Repetitive development tasks

Jenkins is widely used for implementing **CI/CD**.

Why Jenkins?

- Reduces manual work.
- Saves time.
- Automates repetitive tasks.
- Helps detect errors early.
- Supports CI/CD.
- Integrates with tools such as Git, GitHub, Docker, Maven, AWS, and Kubernetes.

Basic Jenkins Components

Jenkins Controller

- Main Jenkins server.
- Manages jobs and pipelines.
- Schedules builds.
- Manages plugins and credentials.

Jenkins Agent

- Machine used to execute Jenkins tasks.
- Performs builds and tests.

Job

- A task configured in Jenkins.
- Example: building an application.

Pipeline

- A sequence of automated stages.

Example:

Checkout → Build → Test → Deploy

Jenkinsfile

- File containing Pipeline instructions.
- Usually stored inside the project's Git repository.

Plugin

- Adds additional functionality to Jenkins.
- Examples: GitHub, Docker, AWS, Maven, and Kubernetes plugins.

Build

- One execution of a Jenkins job or pipeline.

Workspace

- Directory where Jenkins performs its work.

Trigger

- Starts a Jenkins job.
- Examples:
 - Manual trigger
 - GitHub webhook
 - Poll SCM
 - Scheduled trigger

Console Output

- Displays logs generated during a build.
- Helps identify errors and troubleshoot problems.

2. CI/CD Deployment Method Used in Jenkins

Continuous Integration (CI)

Continuous Integration means frequently integrating code changes into a shared repository.

Jenkins can:

1. Detect new code changes.
2. Download the latest code.
3. Build the application.
4. Run tests.
5. Report the result.

This helps identify problems early.

Continuous Delivery

In Continuous Delivery:

- Code is automatically built.
- Tests are automatically performed.
- The application is kept ready for deployment.
- Deployment may require manual approval.

Continuous Deployment

In Continuous Deployment:

- Code is automatically built and tested.
- If the tests are successful, Jenkins automatically deploys the application.

Basic CI/CD Flow

Developer



GitHub



Jenkins



Build



Test



Deploy

Jenkins Pipeline

A Jenkins Pipeline can divide the CI/CD process into different stages.

Example:

Checkout



Build



Test

↓
Deploy

Each stage performs a specific part of the deployment process.

Jenkins with GitHub

A common workflow is:

Developer
↓
Push Code
↓
GitHub
↓
Jenkins
↓
Build & Test
↓
Deploy

Jenkins can be triggered by a GitHub webhook whenever new code is pushed.

3. Installing Jenkins on Local Machine

Jenkins was installed on a **RHEL EC2 instance** for the practical.

3.1 Installing Java

Jenkins requires Java to run.

Java was installed using:

```
sudo dnf install java-21-openjdk -y
```

The Java version was verified using:

```
java -version
```

The installed version was:

OpenJDK version "21.0.12.1" 2026-08-18 LTS

3.2 Installing wget

The wget package was installed because it was required to download the Jenkins repository.

```
sudo dnf install wget -y
```

3.3 Adding Jenkins Repository

The Jenkins repository was downloaded using:

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \  
https://pkg.jenkins.io/redhat-stable/jenkins.repo
```

The Jenkins repository key was imported using:

```
sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2026.key
```

The package information was refreshed using:

```
sudo dnf makecache
```

3.4 Installing Jenkins

Jenkins was installed using:

```
sudo dnf install jenkins -y
```

The installation completed successfully.

3.5 Starting Jenkins

The Jenkins service was started using:

```
sudo systemctl start jenkins
```

The status was checked using:

```
sudo systemctl status jenkins
```

The service showed:

Active: active (running)

```
[ec2-user@ip-172-31-8-1 ~]$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disa>
   Active: active (running) since Wed 2026-09-30 06:44:36 UTC; 17s ago
     Invocation: 0d876e4e6c5646418b76269cf0fab54b
       Main PID: 2144 (java)
         Tasks: 41 (limit: 4156)
        Memory: 387.9M (peak: 399.7M)
          CPU: 19.681s
         CGroup: /system.slice/jenkins.service
                └─2144 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jen>
```

3.6 Enabling Jenkins at Boot

Jenkins was configured to start automatically after a system restart.

```
sudo systemctl enable jenkins
```

The configuration was verified using:

```
sudo systemctl is-enabled jenkins
```

The output was:

```
enabled
```

3.7 Verifying Jenkins Port

Jenkins uses port **8080** by default.

The port was checked using:

```
sudo ss -lntp | grep 8080
```

The output confirmed that Jenkins was listening on port 8080.

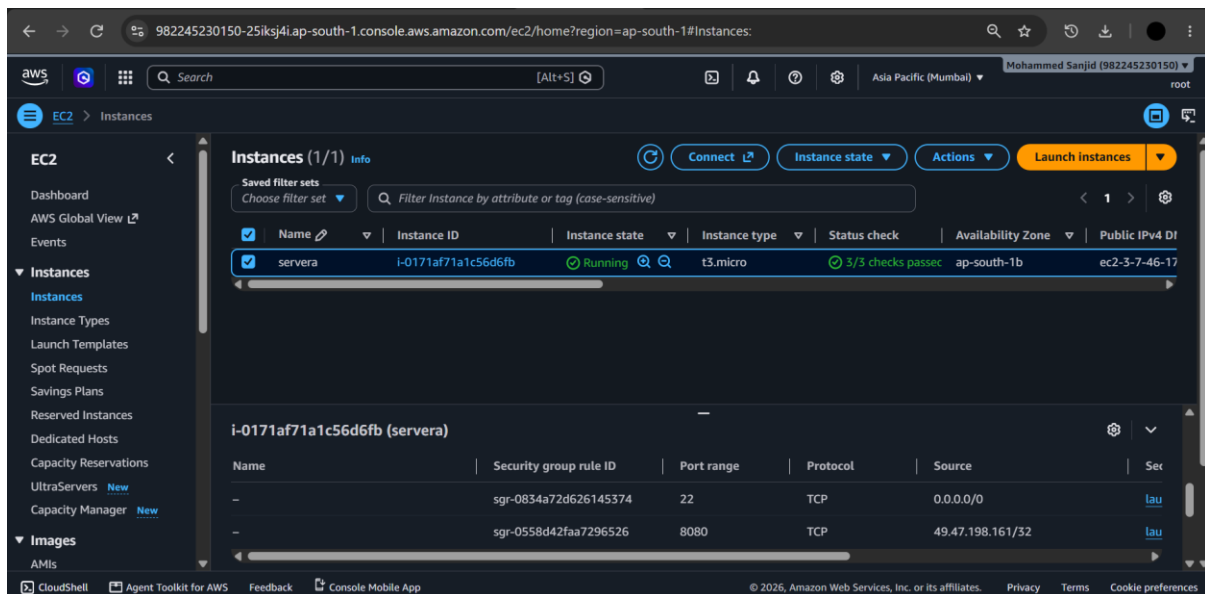
```
[ec2-user@ip-172-31-8-1 ~]$ sudo ss -lntp | grep 8080
LISTEN 0      50          *:8080      *:~        users:((("java",pid=2144,fd=9))
```

3.8 Configuring AWS Security Group

Since Jenkins was installed on an AWS EC2 instance, port **8080** was allowed through the EC2 Security Group.

The inbound rule was configured as:

Type	Protocol	Port	Source
Custom TCP	TCP	8080	My IP



3.9 Accessing Jenkins

Jenkins was accessed using the EC2 public IP address:

<http://<EC2-Public-IP>:8080>

The Jenkins initial setup page was displayed.

3.10 Getting the Initial Administrator Password

The initial Jenkins administrator password was obtained using:

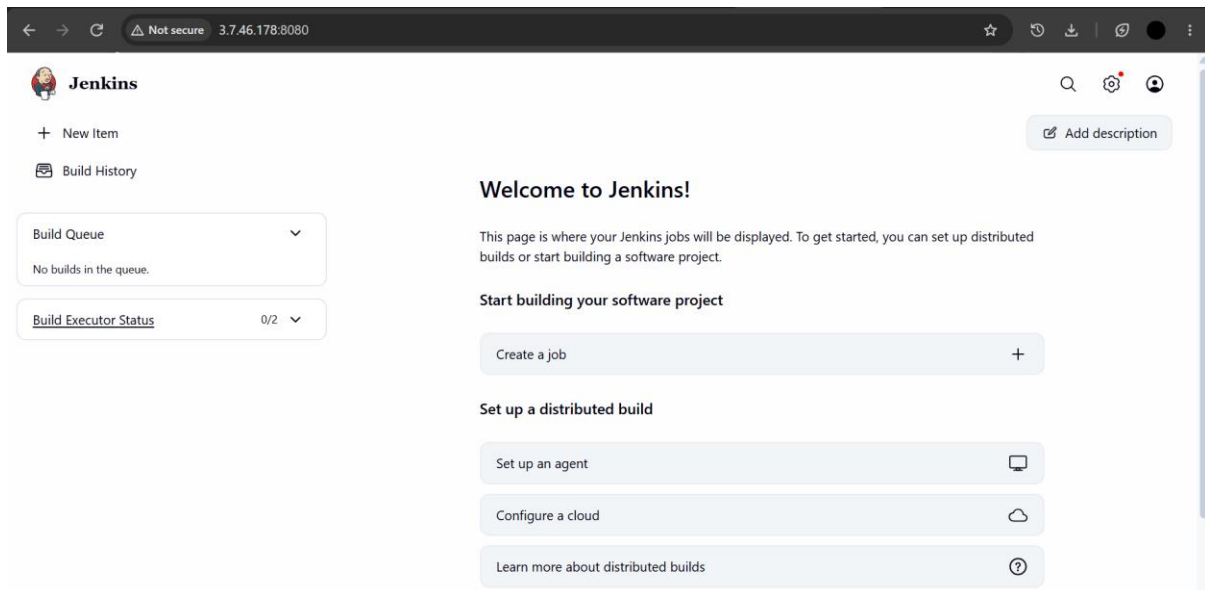
```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

The password was entered into the Jenkins setup page.

3.11 Completing Jenkins Setup

The recommended plugins were installed during the initial setup.

The administrator account was configured, and the **Jenkins Dashboard** was successfully accessed.



Study Notes

Jenkins

- Open-source automation server.
- Used for build, test, and deployment.
- Commonly used with CI/CD.

CI

- Continuous Integration.
- Automatically builds and tests code changes.

CD

- Continuous Delivery or Continuous Deployment.
- Helps automate application delivery and deployment.

Job

- A task performed by Jenkins.

Pipeline

- A sequence of automated stages.

Jenkinsfile

- Contains Pipeline instructions.

Plugin

- Adds extra functionality to Jenkins.

Controller

- Main Jenkins server.

Agent

- Executes Jenkins tasks.

Build

- One execution of a job or pipeline.

Workspace

- Directory where Jenkins performs its work.

Trigger

- Starts a Jenkins job.

Important Jenkins Port

8080

Jenkins URL

<http://3.7.46.178:8080>

Important Commands

```
java -version
```

Checks Java.

```
sudo systemctl start jenkins
```

Starts Jenkins.

```
sudo systemctl status jenkins
```

Checks Jenkins status.

```
sudo systemctl enable jenkins
```

Starts Jenkins automatically after reboot.

```
sudo ss -lntp | grep 8080
```

Checks Jenkins port.

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

Gets the initial Jenkins password.

Conclusion

Jenkins was successfully studied as an automation server, including its basic components, jobs, pipelines, plugins, and workspace. The **CI/CD deployment method** used with Jenkins was also studied, including Continuous Integration, Continuous Delivery, and Continuous Deployment. Finally, Jenkins was successfully installed and configured on the RHEL EC2 instance, and the Jenkins Dashboard was accessed through port 8080.